AI Customer Support Agent — Messaging Architecture

How Twilio (RCS/SMS) talks to your app — and how your AI replies

Read left → right for inbound. Bottom dashed arrow is the reply path.

Inbound = Twilio pushes a webhook to you. Outbound = your app calls Twilio's API.

Local development tip

Your laptop is not on the public internet by default.
Use a secure tunnel (e.g. ngrok) so Twilio can reach:
https://your-tunnel.example → http://localhost:8000/webhooks/twilio
In production, Twilio points at your real HTTPS server URL.

Production habits (do these early)

- Store Twilio credentials in .env — never commit secrets
- Validate Twilio webhook signatures (prove requests are real)
- Log conversation IDs for audit / support review
- Escalate uncertain or high-risk cases to a human

Step-by-step: one customer message

A

Customer sends a message

From their phone over RCS (or SMS if RCS is unavailable).

B

Twilio receives it

Twilio owns carrier delivery. Your app does not talk to carriers directly.

C

Twilio calls your webhook

HTTP POST with From, Body, MessageSid, etc. This is the “push” into your environment.

D

Your API accepts the request

FastAPI endpoint validates signature, parses the payload, starts a conversation turn.

E

AI Agent decides what to do

Look up knowledge, call tools, draft a reply, or escalate to a human.

F

Your app calls Twilio to reply

Outbound REST/SDK send. Twilio delivers RCS/SMS back to the customer.

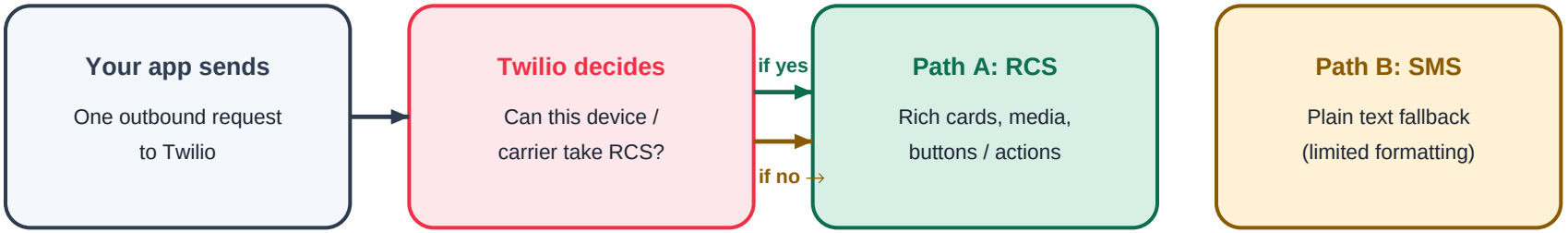
Remember the two directions

Inbound: Twilio → your webhook (they call you)

Outbound: your app → Twilio API (you call them)

RCS → SMS fallback (one page)

RCS is richer messaging. SMS is the reliable backup. Your AI should assume either path can deliver the reply.



Why fallback happens

- Device does not support RCS
- Carrier / region has no RCS path
- Customer has RCS disabled
- Temporary delivery failure on RCS
- Sender / Messaging Service not RCS-ready

What this means for your AI agent

- Prefer clear plain-text answers first
- Treat buttons / rich cards as optional UX
- Never put critical info only in a button
- Keep replies short enough for SMS
- Log channel used (RCS vs SMS) when available

Rule of thumb

Your app usually sends once. Twilio (plus Messaging Service settings) chooses RCS when possible and SMS when not.

Design the support experience so SMS-only customers still get a complete, safe answer — then enhance with RCS when available.